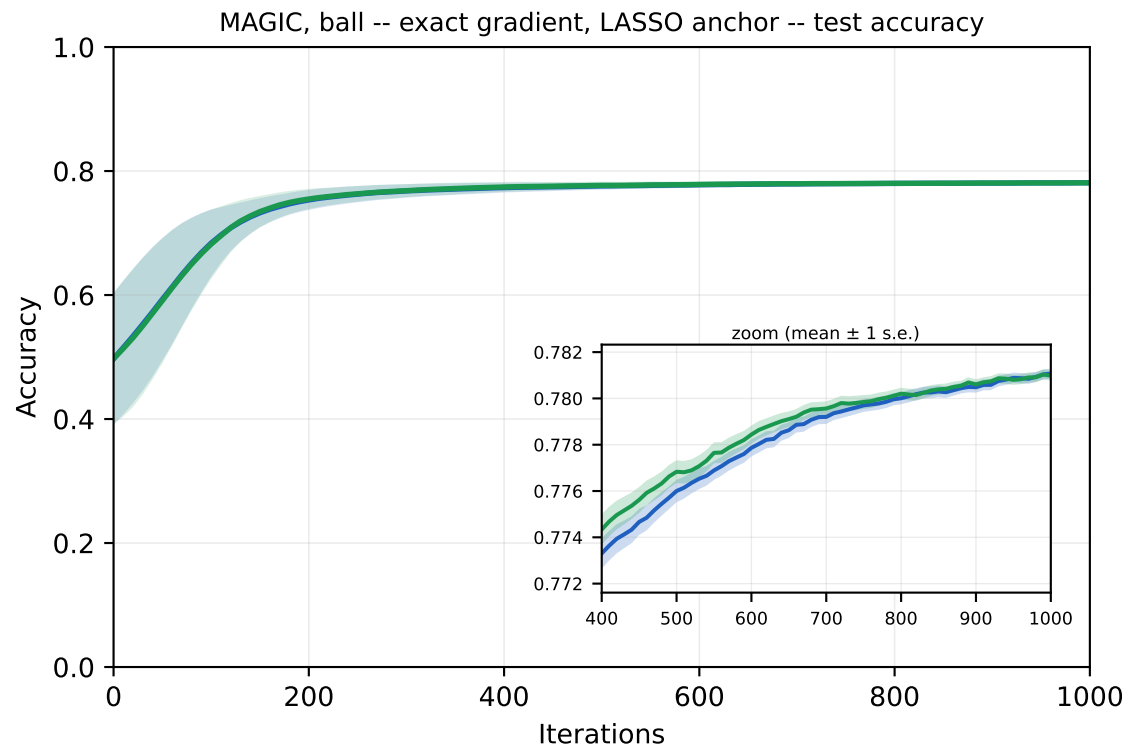
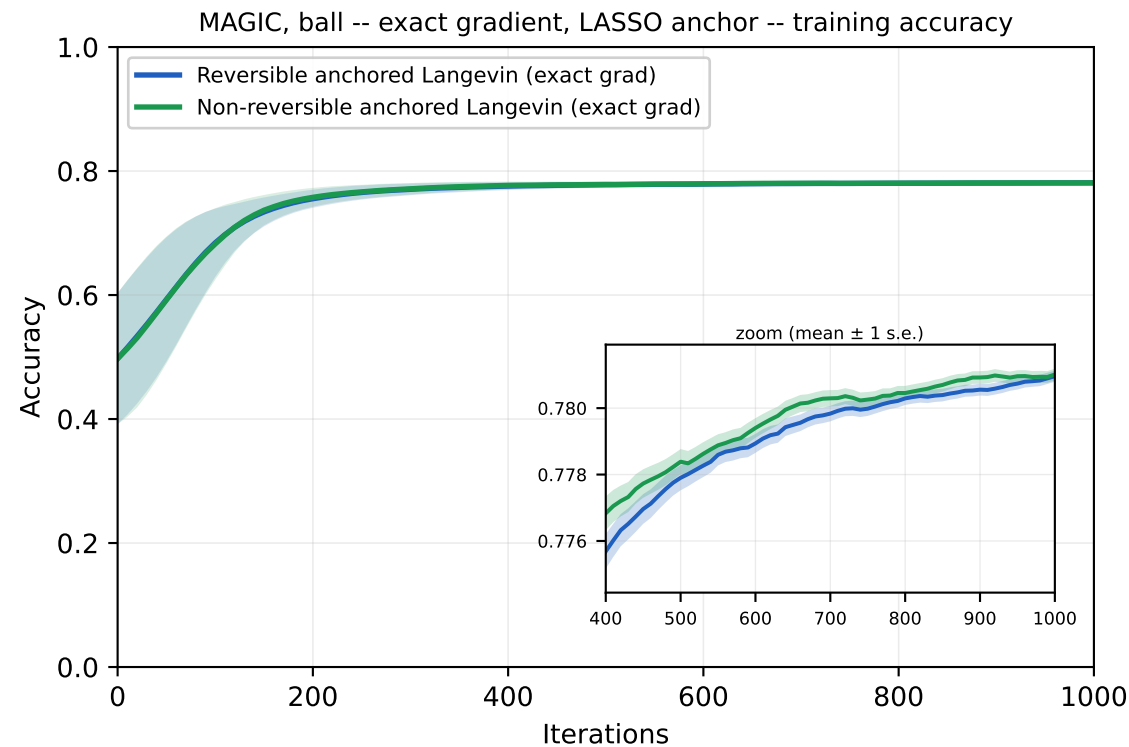


# magic\_ball: exact-gradient anchored Langevin, s = (5, 5, 5)



MAGIC -- ball  $\|w\|^2 \leq 2$  (radius  $\sqrt{2}$ ).  $d = 10$  (intercept  $w_0$  + 9 features).  $U = \sum \text{softplus} - y x \cdot w + w_0^2 / (2 \sigma^2) + \lambda \sum_{j \geq 1} |w_j|$ ;  $U_0$  replaces  $|w_j|$  by  $\sqrt{w_j^2 + \delta^2}$ ;  $a = \exp(U - U_0)$ . EXACT gradient (no mini-batching).  $\sigma = 10$ ,  $\lambda_{\text{lasso}} = 152 = 0.01 n_{\text{train}}$ ,  $\delta = 0.000506$  (so  $a$  in  $[0.50, 1]$ ). Block strengths  $s = (5, 5, 5)$ , acting on  $w_{1..9}$  only. Step  $\eta = 1e-06$  (actually used).  $R = 150$  independent replicates on one fixed stratified 80/20 split ( $n_{\text{train}} = 15216$ ,  $n_{\text{test}} = 3804$ ). Bands: mean  $\pm$  1 SD (ddof = 1) across replicates -- repeat-run variability at a fixed split, not a confidence or credible interval.  $\eta$  and  $s$  selected by a TRAINING-ONLY sweep over 5 step sizes  $\times$  4 strengths; these curves are a CONFIRMATION run on independent replicates (seed 4100) not used for that selection. Verdict on the paired training difference: no significant difference.